

ELK Stack Troubleshooting Guide (Session Summary)

Environment

Filebeat (7.17.29) on application server -> Logstash -> Elasticsearch -> Kibana.

Issues Encountered

1. Filebeat YAML syntax error (mapping values are not allowed). Fixed YAML indentation and validated with **filebeat test config**.
2. Filebeat service failed. Confirmed output using **filebeat test output**.
3. Spring Boot Maven plugin error. Verified pom.xml contained spring-boot-maven-plugin and investigated project structure.
4. Invalid/corrupt JAR. Determined application JAR was unusable and generated test logs manually in app.log.
5. Filebeat harvested no files initially. Verified input path, permissions, registry offsets, and exported configuration.
6. Confirmed Filebeat registry advanced as new log lines were appended.
7. Verified Logstash received events using journalctl and rubydebug output.
8. Confirmed Elasticsearch indexed documents (logs-2026.07.03).
9. Kibana initially failed because .kibana saved-object index was missing. Restarting Kibana recreated required system indices.
10. Discover initially showed no data; verified index existence and advised checking time range and data view.

Useful Commands

Filebeat

```
sudo filebeat test config
sudo filebeat test output
sudo systemctl status filebeat
sudo journalctl -u filebeat -f
```

Logstash

```
sudo systemctl status logstash
sudo journalctl -u logstash -f
```

Elasticsearch

```
curl http://localhost:9200/_cat/indices?v
curl http://localhost:9200/_cluster/health?pretty
curl http://localhost:9200/logs-2026.07.03/_search?pretty
```

Kibana

```
curl http://localhost:5601/api/status
```

Key Lessons

- Always validate Filebeat configuration before restarting the service.
- Confirm Filebeat → Logstash connectivity before checking Elasticsearch.
- Use Logstash logs to verify incoming events.
- Verify indices exist in Elasticsearch before troubleshooting Kibana.
- If Kibana reports missing .kibana indices, restart Kibana and verify system indices are recreated.
- In Discover, ensure the correct data view (logs-*) and an appropriate time range are selected.

End-to-End Validation

1. Append a line to app.log.
2. Verify Filebeat harvests it.
3. Verify Logstash receives it.
4. Verify Elasticsearch document count increases.
5. Refresh Discover in Kibana.